

Vector calculus, without the finger-typing

The gradient of a scalar field φ :

$$\nabla\varphi = \frac{\partial\varphi}{\partial x}\hat{x} + \frac{\partial\varphi}{\partial y}\hat{y} + \frac{\partial\varphi}{\partial z}\hat{z}$$

A total derivative and Gauss's law, in notation the package builds for you:

$$\frac{df}{dt} \quad \nabla \cdot \boldsymbol{E} = \frac{\rho}{\varepsilon_0}$$